

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Chemical Reactions and Equations

CBSE Science 2026-27 | Board-examinable

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

Balance: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$. Name the reaction pattern.

Answer:

$3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Iron displaces hydrogen from steam, so it is a displacement reaction; iron is oxidised and water is reduced.

Question 2

Why should a chemical equation be balanced?

Answer:

Atoms are neither created nor destroyed in an ordinary chemical reaction. Balancing makes the number of each type of atom equal on both sides and expresses conservation of mass.

Question 3

What is observed when iron nails are placed in copper sulphate solution? Explain.

Answer:

A reddish-brown copper coating appears on the nails and the blue solution turns greenish. More reactive iron displaces copper: $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$.

Question 4

What is the central idea or skill tested in Chemical Reactions and Equations?

Answer:

The central focus is: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Chemical Reactions and Equations.

Answer:

First point: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Second point: Atoms are neither created nor destroyed in an ordinary chemical reaction. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Chemical Reactions and Equations lead to an important result or conclusion?

Answer:

Begin with the key idea: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Then connect it to the result: Atoms are neither created nor destroyed in an ordinary chemical reaction.

Chemical Reactions and Equations - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Chemical Reactions and Equations?

Answer:

Use this supporting detail: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Iron displaces hydrogen from steam, so it is a displacement reaction; iron is oxidised and water is reduced. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Chemical Reactions and Equations.

Answer:

Atoms are neither created nor destroyed in an ordinary chemical reaction. Balancing makes the number of each type of atom equal on both sides and expresses conservation of mass. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Chemical Reactions and Equations?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: A reddish-brown copper coating appears on the nails and the blue solution turns greenish.

Question 10

Compare the first and second important ideas in Chemical Reactions and Equations.

Answer:

The first idea is: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. The second idea is: Atoms are neither created nor destroyed in an ordinary chemical reaction. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Chemical Reactions and Equations.

Answer:

Cause or starting point: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Effect or outcome: Atoms are neither created nor destroyed in an ordinary chemical reaction. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Chemical Reactions and Equations?

Answer:

A clear sequence is: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Next, Atoms are neither created nor destroyed in an ordinary chemical reaction. Finally, A reddish-brown copper coating appears on the nails and the blue solution turns greenish.

Chemical Reactions and Equations - Continued

Question 13

Give a 40-50 word answer outline for an important question from Chemical Reactions and Equations.

Answer:

Start with the direct answer: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Add one supporting point: Atoms are neither created nor destroyed in an ordinary chemical reaction. End with the outcome or message: A reddish-brown copper coating appears on the nails and the blue solution turns greenish.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Chemical Reactions and Equations?

Answer:

Use an accurate scientific statement: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Chemical Reactions and Equations?

Answer:

Connect the idea to its reason or result. For instance: Atoms are neither created nor destroyed in an ordinary chemical reaction. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Chemical Reactions and Equations.

Answer:

A sound inference is that a reddish-brown copper coating appears on the nails and the blue solution turns greenish. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Chemical Reactions and Equations.

Answer:

Apply the concept by first stating: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Then justify the response with: Atoms are neither created nor destroyed in an ordinary chemical reaction.

Question 18

What would make an answer from Chemical Reactions and Equations incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Also add: A reddish-brown copper coating appears on the nails and the blue solution turns greenish.

Chemical Reactions and Equations - Continued

Question 19

Prepare an Assertion-Reason style explanation based on Chemical Reactions and Equations.

Answer:

Assertion: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Reason: Atoms are neither created nor destroyed in an ordinary chemical reaction. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Chemical Reactions and Equations in two clear steps?

Answer:

Step 1: state the concept or event - $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Step 2: support it with the explanation or result - Atoms are neither created nor destroyed in an ordinary chemical reaction.

Question 21

What answer structure is suitable for a 3-mark question from Chemical Reactions and Equations?

Answer:

Use three connected points: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Atoms are neither created nor destroyed in an ordinary chemical reaction. A reddish-brown copper coating appears on the nails and the blue solution turns greenish. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Chemical Reactions and Equations?

Answer:

Give an introduction, develop these key points - $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Atoms are neither created nor destroyed in an ordinary chemical reaction. - and close with: A reddish-brown copper coating appears on the nails and the blue solution turns greenish. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in Chemical Reactions and Equations be applied to a new situation or case study?

Answer:

Identify the relevant principle first: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Then use the chapter's reasoning to explain the case: Atoms are neither created nor destroyed in an ordinary chemical reaction.

Question 24

Which detail from Chemical Reactions and Equations is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: A reddish-brown copper coating appears on the nails and the blue solution turns greenish. Explain it in context instead of using it as an isolated statement.

Chemical Reactions and Equations - Continued

Question 25

Write a brief concluding statement for an answer on Chemical Reactions and Equations.

Answer:

A suitable conclusion is: A reddish-brown copper coating appears on the nails and the blue solution turns greenish. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for Chemical Reactions and Equations.

Answer:

1. $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. 2. Atoms are neither created nor destroyed in an ordinary chemical reaction. 3. A reddish-brown copper coating appears on the nails and the blue solution turns greenish. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Chemical Reactions and Equations before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. - and supported it with a reason, example, calculation, or outcome such as: Atoms are neither created nor destroyed in an ordinary chemical reaction.

Question 28

Why does Chemical Reactions and Equations remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Atoms are neither created nor destroyed in an ordinary chemical reaction. A reddish-brown copper coating appears on the nails and the blue solution turns greenish.

Question 29

Create a final 60-80 word answer plan for a long question from Chemical Reactions and Equations.

Answer:

Opening: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. Development: Atoms are neither created nor destroyed in an ordinary chemical reaction. Conclusion: A reddish-brown copper coating appears on the nails and the blue solution turns greenish. Use complete sentences and include all required evidence, steps, or references.

Question 30

What is one last-minute revision question you should ask yourself about Chemical Reactions and Equations?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$. and Atoms are neither created nor destroyed in an ordinary chemical reaction.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.